State empathy in the context of personality assessment through virtual reality and nonverbal items

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ABSTRACT

Psychological assessment is a construct often used in various domains, from the organizational field, to the educational or the clinical one, having strong implications at the level of job performance, of psychotherapy, or of the choices regarding the career path. For this reason, the identification and the understanding of the variables that might influence this process is crucial. The current study had as main objective the analysis of state empathy in the context of personality assessment in two forms – with items that were presented in the virtual reality environment, respectively as illustrations. The focus was on its components and on the relationship between them, but also on the link between each of them with the measured personality facets. The main results showed a higher level of state empathy when measured after the VR assessment, but also a significant relationship between this and empathy as a general trait. When it comes to the interactions with the personality dimensions, the results showed significant and positive correlations with Openness to feelings, Altruism and Self-discipline, and negative with Antagonism. This study makes a valuable contribution to the specialized literature where the relation between state empathy and personality is poorly investigated, especially under the context of virtual reality or non-verbal items.

Key words: state empathy; general trait empathy; personality assessment; virtual reality environment; non-verbal items.

INTRODUCTION

Empathy – the general construct

Empathy is a concept in psychology that can be described from several perspectives, being known for its "multifaceted" dimension. Most studies agree that empathy can be divided into two components, namely cognitive and affective. The first of them refers to the way in which an individual manages to understand, from an objective point of view, the emotions of the other. In other words, this facet refers to the idea of understanding a perspective separate from one's own. The affective one, however, considers a person's capacity to feel on one's own skin the emotions experienced by the person next to them, their direct appropriation (Larionow and Preece , 2023). This differentiation is important and it is suggested to be considered when measuring empathy, but especially when it comes to understanding it and how it influences individuals and the social contexts in which they find themselves.

Cognitive empathy, and its separation from emotional empathy, can be understood through its association with Theory of Mind. This refers to how an individual is aware that those around them have a set of beliefs, feelings, and emotions separate from their own, and can understand and often predict them, depending on the context in which they find themselves. A promising context for studying this relationship is that of autism spectrum disorder (ASD), which is often associated with an inability to use Theory of Mind (Baron-Cohen, 1991, Ozonoff et al., 1991). Thus, a study conducted by Dziobek et al. used the Multidimensional Empathy Test on a group of individuals diagnosed with Asperger syndrome. The results showed that the participants showed a normal level of emotional empathy, but had a clear deficit in objective understanding of the state of others. Similarly, it has been observed that people with autism show normal brain activity when looking at people in pain, having this contagion of pain, but when it comes to cognitive empathy, there are clear deficiencies (Hadjikhani et al., 2014). Thus, by analyzing people with ASD, we can understand the clear differentiation between the 2 types of empathy and the importance of separating them in the context of research (Calderoni et al., 2013, Greenberg et al., 2018).

An important perspective on empathy is also given by the field of neuroscience , by exploring brain activity at the moment when we would believe that this phenomenon is present. Perhaps one of the most interesting results is that when observing a person experiencing a certain emotion, certain regions are activated in the neural plane that are also active when the individual feels that emotion himself. Among these regions are the anterior part of the insula and the anterior middle cingulate cortex, thus resulting in the theory of "shared representation", which assumes this overlap between the two experiences, confirming the existence of affective empathy (Bernhardt and Singer, 2012, Lamm et al., 2011, Tully and Petrinovic, 2017). By analyzing empathy as a neurological phenomenon, the separation between the two components can also be highlighted. By focusing attention on certain lesions in the brain, it was observed that the inferior frontal gyrus influences only affective empathy, while damage to the ventromedial prefrontal cortex leads to deficiencies in the manifestation of cognitive empathy (Perry et al., 2017). Thus, the two concepts are determined at the cerebral level by different systems, but still interconnected, highlighting again the need, but also the benefits brought by studying them individually, without neglecting the relationship between them.

Although empathy is a universal concept, it is crucial to also have a perspective on the differences that may appear in its manifestation due to cultural differences. Being a phenomenon that develops, but also deeply influences social relationships, it is understandable that the environment in which an individual grows up is an element to be

taken into account. Studies show that both perspectives – affective and cognitive – have an influence from a cultural point of view. For example, from the point of view of cognitive empathy, the difference appears in the way of thinking and relating to others when an individual comes from an individualistic society, versus from a collectively oriented one. Thus, for the former, it is characteristic of individuals to have a vision focused on personal autonomy, different from the latter, in which members will be more directed towards social harmony, cognitive empathy having a higher level in the case of those coming from a culture defined by sociocentrism. This type of difference was not found, however, when it comes to affective empathy (Jami et al., 2018). It is also worth noting that, especially in intercultural contexts, higher intercultural intelligence leads to a higher level of cognitive empathy (Kong et al., 2020). When it comes to affective empathy, however, there are also studies that have found that it falls under the influence of cultural differences – in the case of Westerners, for example, who showed a higher level in this regard, compared to members of Eastern cultures (Atkins et al., 2016). From a gender perspective, both in children and in adulthood, some studies show that girls show a higher level of empathy, from the point of view of both facets (Michalska et al., 2013, Cherewick et al., 2022). This makes sense, given society's tendency to stop boys from expressing themselves emotionally from an early age (Topçu and Erdur - Baker, 2012, Lam et al, 2012). Another differentiation between the components of empathy, which can help us understand empathy and its importance, is given by the analysis from the perspective of age. Thus, people who are older tend to lose the capacity for cognitive empathy due to the degradation of certain regions of the brain, while affective empathy remains constant or even shows an increase, possibly due to the unfulfilled need of older people for emotional connection (Jarvis et al., 2024, Beadle and Vega, 2019), and from a neural perspective, the activation pathway differs from younger individuals (Moore et al., 2015).

Empathy as a state and empathy as a general trait

The multidimensional nature of empathy is not limited to the cognitive and affective distinction. Another dual way of viewing this psychological phenomenon is by separating empathy as a general trait and empathy as a state. The former can be conceptually understood as a characteristic of an individual’s personality, a component that remains constant, unaffected by context, indicating the ability to rationalize the perspectives of others, but also to feel their emotions. It is generally measured by self-assessment, making it more difficult to minimize the impact of the bias of being socially desirable (Murphy and Lilienfeld, 2019). In contrast, empathy as a state refers to the response that an individual has when placed in contexts that are supposed to generate empathy – observing an experience that a peer is going through. Thus, this type of empathy is clearly influenced by the context, being described by the immediate, possibly fluctuating reaction of the individual to various stimuli. The measurement consists of assessing the actual reactions of the participants, thus being less susceptible to the bias, the assessment having a more objective note (Crocetti et al., 2016).

Social behavior and interpersonal relationships are concepts often seen as closely related to empathy. A study by Eisenberg et al. in 2010 found that people who scored higher on trait empathy were more likely to engage in altruistic actions. In the context of a group, solidarity and togetherness were found to be stronger when members had a higher empathic connection, suggesting that the benefits of empathy extend beyond the individual level (Eisenberg et al., 2010). Furthermore, this idea continues at the societal level, with higher general empathic tendencies being correlated with a sense of collectivism and kindness (Chopik et al., 2016). If we look at empathy as a state, however, situational factors or a

certain perception of social norms can both positively and negatively influence the relationship between empathy and prosocial behavior (Kira et al., 2021).

The possibility of distinguishing between the cognitive versus affective facet also extends to the two types of empathy, leading to a deeper and more complex understanding.

The cognitive component in the case of empathy as a trait influences how accurately an individual can predict the feelings or behavior of the person next to him, but also how good his understanding of social norms is, which facilitates the existence of constructive interpersonal relationships. For example, the moderating role of this component has been studied in the context of aggressive responses to certain stimuli in relation to general traits of anger, resulting in the fact that a higher level facilitates the regulation of one's own emotions, since by taking into account the perspective of the other, the need to manifest anger decreases (Qi et al., 2019). However, if we look from the perspective of the affective component, the results vary. In certain contexts, a higher level increases the presence of prosocial behavior, due to stronger feelings of compassion, but there have also been cases in which, because personal discomfort was in turn felt more strongly, aggression was also increased (Pang et al., 2022).

Empathy as a state also takes on a more dynamic form by considering the two concepts, simultaneously focusing on what an individual thinks at the moment from the perspective of a peer, but also what emotions they experience at the moment when they observe the behavior of the other. Compared to empathy as a trait, which has a more universal character, the state one is easier to manipulate. Following a longitudinal study, it was demonstrated that both from an affective and cognitive point of view, empathy as a state increases significantly following active interaction with social media (Vossen and Valkenburg, 2016). In addition, in the context of reducing stigma against people suffering from HIV/AIDS, the level of cognitive empathy as a state increased after reading neutral news related to scientific research in the field, which led to an increase in empathy as a state itself, due to the elimination of the participants' preconceptions towards this disease (He et al., 2022). Thus, the idea that empathy as a state can be more easily induced and regulated in experimental contexts is reinforced, providing an effective framework for exploring how it influences prosocial behaviors and attitudes. Also, a study conducted in 2016 by Van der Graaff et al. analyzed the two types of empathy in relation to motor empathy, which refers to the physical reactions that an individual has in relation to another's experience, along with the tendency to mirror the facial expressions of others, being seen as a fairly automated process, lacking a cognitive essence (Hoffman, 1984). The findings show that in the case of happiness there are differences given by the gender of the participants, with cognitive state empathy being significantly predicted by motor empathy only in the case of boys, but affective empathy had similar results, regardless of gender. In the case of sadness, motor empathy continued to be a significant predictor of affective state empathy, the latter becoming a mediator in the relationship between cognitive state empathy and motor empathy (Van der Graaff et al., 2016). In addition to these two dimensions discussed extensively, especially regarding empathy as a trait, and then as a state, another one is defined, in the context of empathy as a state: associative. Although there are fewer studies that focus on this component, it comes from the foundations of the concept of empathy, namely inclusive fitness and reciprocal altruism (Preston & de Waal , 2002, cf. Shen , 2010), which come from evolutionary psychology and refer to the adaptive function of empathy to facilitate cooperation and survival within a species by emphasizing the building of stable interpersonal relationships. This can also be called “identification” (Campbell & Babrow, 2004; Chory - Assad & Cicchirillo , 2005; see also Decety & Jackson, 2006, cf. Shen, 2010), referring to the transposition of oneself completely into a presented situation, as if the individual were him/herself instead of the depicted character. It is the component most closely related to the

development of relationships (Davis, 1994, cf. Shen, 2010), social cohesion increasing if there is a feeling of identification with those around him/her. It often appears and is measured in the context of media consumption, films, videos, books or stories. Thus, it refers to the indirect experience of the presented situation, both at an emotional and cognitive level (without overlapping with the other two components), being "the link between perception and action in the process of empathy as a state" (Shen, 2010). The conclusions we can draw from this research are that considering empathy as a multifaceted model is very important, leading to complex results that help us understand how the dimensions relate to each other, and what effect they have on social behavior.

One of the most important and interesting aspects to look at when we talk about the distinction between trait versus state empathy is the relationship between the two, which can also include the cognitive and affective divide. It is quite natural to assume that people who have high levels of trait empathy will also score higher on state empathy. There are numerous studies that have found a positive relationship between the two types of empathy, but only considering the affective level for both (Van der Graaff et al., 2016, cf. Batson et al., 1986; Eisenberg , Miller, et al., 1989; Eisenberg et al., 1988; Eisenberg et al., 1994; Westbury and Neumann, 2008; but see Eisenberg et al., 1991). In contrast, the link between affective state empathy and cognitive trait empathy differs from one study to another, with findings both validating it as significant and denying this aspect, or in which the relationship is positive, but either only in the case of girls, or in the context in which participants were explicitly told to try to evoke empathy (Van der Graaff et al., 2016, cf. Eisenberg et al., 1988; Eisenberg et al., 1994; Eisenberg , Miller, et al., 1989; Eisenberg et al., 1991). The results of the study conducted by Van der Graaff et al. are slightly different from expectations, since, in the case of sadness for example, although the associations between trait empathy and state empathy, when the correspondence between the affective and cognitive components was made, were positive, their magnitude was low. Regarding happiness, the results are similar for affective empathy, but cognitive empathy as a trait was found to have a significant relationship with affective empathy as a state only in the case of girls. Another study, however, showed that exposure to intense social exclusion has a strong effect on the relationship between the two types of empathy. Thus, in the context of people for whom social exclusion had a lower degree, state empathy proved to be a mediator in the relationship between low aggressive tendencies and trait empathy. However, mediation no longer occurred when individuals were exposed to intense social exclusion for a long time, a possible argument being that they reach an "emotional numbness", which results in a decrease in the level of empathy as a state. In such cases, the relationship between the types of empathy is no longer significant, with high aggressive tendencies appearing even in the case of participants who obtained high scores in the measurement of empathy as a trait (Lu et al., 2022). The absence of a significant interaction between the two specified dimensions is also highlighted by the research of Wood et al. from 2013. They analyzed the effects of empathy on the attribution of remorse to the defendant in the context of a legal decision, and empathy as a trait did not prove to be a moderator of the effects of empathy, nor vice versa. The result seemed counterintuitive, given that previous studies have shown that the two dimensions often interact (Wood et al., 2013 cf. Rusting, 1999). However, in analyzing how individuals with different levels of trait empathy respond to facial expressions measured by event-evoked potentials (ERPs), the results indicated the existence of an interaction. Attention to the expressions of others was higher in participants with a high level of trait empathy, indicating that they are more likely to show responses that require momentary emotional understanding, such as distinguishing between happy and angry faces, where state empathy is also high (Choi & Watanuki, 2014). Therefore, following the major differences from one study to another regarding the connection between the two forms of empathy, it can be clearly said that more research is

needed in the field. It can also be deduced that an essential factor to consider is the contextual reality in which the studies are conducted, which reinforces the need to consider empathy as a state as a separate variable in the process of analyzing empathy as a psychological phenomenon.

The differences between women and men are also a point to consider from the perspective of separating the two dimensions of empathy, with studies differing depending on the situation in this case as well. By referring to empathy as a state felt following short stories (in English, “vignettes”) with heterosexual and homosexual people, respectively, it was noted that the overall score obtained by women was higher, especially in cases where the emotional valence of the situations presented was negative. At the same time, empathy as a trait was also taken into account, and the average score was higher in the case of women. At the same time, the interaction between the two forms proved significant, regardless of gender (Winward , & Itier , 2025). Similar results were also obtained in a study that also took into account the effect of culture on these interactions. Trait empathy was found to be higher in women, but this effect was considerably greater in Western cultures than in Eastern cultures, where the differences given by the gender of the participants were more flattened. For empathy as a state, things become much more complex, and the interaction between gender, culture, and empathy is less clear. The results varied greatly depending on gender, ethnicity, but also on the situation that was presented to the participants, in terms of emotional valence, but also on the gender or culture of the people in the situations they had to analyze. Therefore, this differentiation between Eastern and Western cultures no longer appeared, and from a gender point of view, women obtained a significantly higher score only in the case of cognitive empathy, only in the case of one of the two tasks that the participants had to perform (Zhao et al., 2021). Results that were contrary to expectations were also obtained by Wood et al., in 2013. Although, again, women were found to have higher scores on empathy as a general trait, men were the ones with the higher level of empathy as a state, the authors assuming that this was generated by the fact that in the analyzed situation, the protagonist was male. The general conclusion is that empathy is a very complex, dynamic phenomenon, social behavior depending on the beliefs of the society to which individuals belong (evidence showing that women do indeed have a higher level of empathy as a trait), but situational factors also play an important role, making it impossible to predict people's decisions or attitudes with certainty.

The context of virtual reality. The importance of empathy within it

In a world where the pace of technological evolution has reached a colossal speed, it is normal that psychology will increasingly intervene, leading to new approaches and a more diverse vision of existing issues. The natural question that arises is: "how does this contingency with technology influence individuals, in a field that is entirely with and about people?" The issue comes from two perspectives - how can we use this tool to facilitate what was already known, but perhaps even more importantly, how do these methods influence the people involved and the final result. We need to do this type of research constantly and preventively, in order to be able to control the inevitable impact of these innovations that continue to appear at an increasingly accelerated pace. In this way, having tangible results related to the technologies already present, we can make clearer decisions about what is worth investing effort and resources in, while at the same time anticipating certain problems that may arise in the future.

A context that clearly fits the description above is the use of virtual reality to create new tools that can be used in contexts of psychological interest. And an even more important point than the creation of such a tool itself is the research on the impact it has on people and

the understanding of why that tool works or does not work. Strictly from the perspective of the present study, virtual reality was the pillar of the establishment of a personality assessment system, which wants to bring an innovative touch in the context of assessment and individual differences. Continuing the idea outlined so far, this study does not want to explain in detail how this tool was built, but is based on the analysis of what influences the result to be better than a classic test, but also why it happens. One possible aspect is the empathy felt by participants strictly during the testing, that is, state empathy, moderated by empathy as a general trait of the individual. Why would exploring empathy in this context be important? The basic idea is that through this technological tool, reality is simulated, and the participating individual would be more involved the more authentic the experience is. And one way in which a situation often feels more real is by experiencing and sharing common states and emotions with other people. In the context of this questionnaire, understanding the characters presented and the situations depicted is a key point in completing it as accurately as possible, being able to intuitively compare this idea with the way in which an individual puts themselves in the shoes of another to understand a certain situation, which shows empathy – both at the moment, but also being influenced by how empathetic the person is in general.

Personality testing

Personality testing has been one of the most important and most studied aspects of psychology, both in the early days of this field and now, in modern psychology. Through it, we can understand what influences an individual's thoughts, values, attitudes, we can try to predict behavior in various contexts. Its usefulness is vast, having a great impact in the organizational field, education, therapy and beyond. In this sense, it goes without saying that research and studies on this aspect are vast and dynamic, but also continuous, with new theories or technologies emerging that lead to its improvement.

In the organizational context, personality testing plays a major role as effectively identifying differences between individuals and intuiting certain behaviors is essential in achieving the highest possible performance – in short, performance meaning the extent to which an individual manages to fulfill his or her duties and bring a plus to the organization. Integrating this aspect leads to more objective and more specific hiring decisions for the respective job, to better team cohesion and dynamics, to openness to employee development, as there is a clearer alignment between the characteristics of individuals and the duties, but also the culture of the organization. When it comes to the models that exist and have been empirically tested numerous times, one of the best known and with remarkable results is the Big Five , which describes personality from the perspective of five dimensions: openness to experience, conscientiousness, extraversion , agreeableness and neuroticism . Each of these dimensions has a specific role in predicting performance at work, being able to look at it from different perspectives. Openness to experience refers, as the name suggests, to an individual's willingness to try new things, listen to innovative ideas, and accept variety. The correlation with overall job performance is moderate, depending largely on the type of job and what it entails - more specifically, if it is more complex, unstructured, inclined to creativity, the relevance increases. It has been shown to be a valid criterion only for one of the definitive criteria of performance, training competence, being insignificant for professional competence or personnel data. Thus, individuals who score higher on this factor are more likely to benefit from development or training programs, being more open and motivated to learn new things (Barrick & Mount , 1991). A high level of conscientiousness generally describes a person who takes their job seriously, who can be trusted, who completes their tasks on time and carefully. A meta-analysis by Judge and Ilies (2002) concluded that this factor has a

significant relationship with three central dimensions of motivation – goal setting, expectancy-based motivation, and self-efficacy – and is one of the most important predictors of performance. Extraversion refers to a person's tendency to engage in social interactions, conversations, and being more assertive and energetic. Similar to openness to experience, the relationship with performance varies depending on the type of task, being particularly relevant when the role is characterized by collaboration or interaction with other people (Loveland et al., 2005). In addition, it has been observed that in relation to individual performance within a team, extraversion is a key element, with people with higher scores having a greater impact on team dynamics and being viewed more favorably by colleagues (Barry and Stewart, 1997). Agreeableness is a personality dimension that considers characteristics such as altruism, empathy, kindness, cooperation, with people with higher levels being seen in society as friendly, outgoing, and focused on the needs of others. Studies have shown that this facet has the greatest impact on job satisfaction, which leads to motivation and performance, which probably stems from the individual's nature to reduce conflict, being a good collaborator in teams (Chandrasekara , 2018). Although the positive influence on working with other people is maintained, in certain contexts, being analyzed individually, agreeableness did not have a strong impact on performance (Kluemper et al., 2015). The fifth factor, neuroticism , is the only one that is negatively correlated with employee performance. A higher score refers to the tendency of individuals to experience negative emotions, being more insecure, easily irritable and of increased emotional instability, being more prone to perceive situations in a threatening way and to have a response strongly influenced by emotions. Thus, by reporting to various organizational environments, it has been shown that a high degree of neuroticism is associated with lower satisfaction and efficiency at work, and, overall, with lower performance (Judge et al., 2002). The context is relevant in this case as well, with certain studies showing that the impact of this facet is attenuated when the job offers autonomy and support (Uppal , 2014). In addition, along with conscientiousness, neuroticism has the greatest impact on performance at work, having in turn a significant relationship with all three factors of motivation mentioned above (Judge and Ilies, 2002).

Considering the described dynamics between the five dimensions of personality and the performance of individuals, it becomes quite clear to visualize how psychological testing using the Big Five model leads to favorable results when it comes to the personnel recruitment process. By considering the specific influence of each factor on the performance components, decisions can be made based on the characteristics of the job, so that the individual's fit with the position is maximized.

Empathy and personality dimensions

Empathy as a general trait has been studied in relation to all five dimensions of the model. The one with which it has been shown to have the strongest relationship is agreeableness , which is quite intuitive considering what it implies. Thus, individuals with higher scores on this facet tend to be more attentive and understanding of the emotional state of others, and it is easier for them to see things from another person's perspective (Taylor et al., 2010, Bętkowska-Korpała et al., 2022). Openness to experience is also positively related to empathy, with researchers stating that the explanation lies in the nature of individuals to be more open to diversity, more curious and with a richer imagination. These things are reflected especially on the cognitive dimension, the emphasis not necessarily being placed on directly feeling what others are feeling, but on being open to multiple perspectives and points of view (Song & Shi , 2017; Bętkowska-Korpała et al., 2022). In the case of neuroticism , however, most studies show that the relationship is inverse, especially when it comes to strong,

negative valenced experiences of others. This is caused by the feeling of personal discomfort that occurs when an individual comes into contact with another person's emotions, the ability to relate and understand empathically decreasing the higher the level of neuroticism (Song & Shi , 2017; Costa et al., 2014; Guilera et al., 2019). Extraversion has also been studied in relation to empathy, with most results showing a positive relationship, with higher scores indicating greater sociability and expressiveness (Warrier et al., 2018). In addition, these individuals tend to believe that they are more empathetic, and more casual exposure to social contexts leads to more opportunities to exercise empathy, increasing their capacity for emotional understanding (Giunta et al., 2010). However, certain personality tendencies such as narcissism can intervene in this relationship and reverse it (Ramos- Villagrasa et al., 2020). For the last of the five facets, conscientiousness, its relationship with empathy is significant and positive according to most research. A study by Liaqat in 2022 even finds this facet to be dominant alongside agreeableness in predicting empathy among students in several countries, due to their more diligent and organized style. In the case of medical students for example, where a higher level of conscientiousness means a higher desire to successfully complete tasks, this relationship is again found to be strong (Airagnes et al., 2021). However, there are also studies that do not have similar results, saying that this link is insignificant (O'Tuathaigh et al., 2019 ; Bętkowska-Korpała et al., 2022).

Clinical dimensions of personality, their relationship with performance and empathy

In addition to this model of personality, which depicts the normal characteristics, on a spectrum, of an individual, there are also conceptualized clinical dimensions, for example in the Alternative Model of Personality Disorders, within which three components are distinguished.

The first of them is Negative Affectivity, which can be seen as a composition between Emotional Lability and Hostility. As a whole, it refers to the predisposition of an individual to have negative emotional states, but which fall within the clinical symptomatology – depression, anxiety, general disturbance. It is generally seen as affecting interpersonal relationships, which also leads to having a negative effect on performance at work (Grijalva & Newman, 2014). This fact is also due to the fact that the individual has a lower resistance to stress, being able to give in easily (Genoud & Waroux , 2021). In relation to empathy, the association between the two is a negative one, a high level of this facet becoming a barrier in being aware of the emotional state of others and in providing an adequate response. This link was observed in both cognitive and affective empathy, with the limitation appearing in various contexts in which it was tested (Löchner et al., 2023).

Antagonism is seen in the literature as the opposite of Agreeableness , thus representing an individual's tendency to be argumentative, vindictive, and not to respect social norms. The components that form this dimension are Manipulation Tendency and Affective Insensitivity, and a high level of these negatively affects interpersonal relationships (Venema et al., 2021). Workplace performance is negatively affected overall, as these individuals find it difficult to work in a team and have a tendency to enter into conflicts (Servidio et al., 2024). On the other hand, individuals are more competitive, which can lead to a greater effort to complete tasks, but in general the social problems they present counterbalance – there may be short-term gains, but in the long term they negatively influence performance (Venema et al., 2021; , Hyatt et al., 2024). Just as agreeableness is closely related to empathy, but in a positive sense, the association with Antagonism is the opposite, this being reflected both in understanding emotions and in taking the perspective of other people, appearing to be intolerant of those around them (Lamkin et al., 2017).

Disinhibition is the third clinical dimension and refers to the reduced capacity for impulse control, being composed of Impulsivity and Risk-Taking. The relationship with performance is greatly influenced by the context, but overall it is a negative one, having unfavorable effects even in tasks where it would initially seem useful. Thus, for cases where increased attention, planning and working memory are required, a higher level of Disinhibition has a negative effect on effectiveness (Jonker et al., 2015). On the other hand, there are also jobs where risk-taking is necessary, such as the military environment, but even in such contexts a too high level can lead to accidents, decreasing performance (Breivik et al., 2015). When it comes to the relationship with empathy, Disinhibition is significantly correlated with the lack of empathic capacity. This is explained by the lack of power to regulate one's own emotions (Saraceno et al., 2018), which is further reflected in the understanding of others, these individuals having an inability to respond appropriately to emotional stimuli from others, which causes them problems in social contexts (Filipčíková et al., 2022).

Thus, although the manifestations of the three dimensions differ, the general dynamics towards empathy are the same, being significantly negatively associated with it.

Virtual reality and psychological testing

Virtual reality (VR) technology represents an important step in immersive technology , being a way to allow people to experience a simulation of reality, easily manipulated to illustrate various situations. VR headsets are most often used to access and explore these simulated environments, by stimulating multiple senses, mainly hearing and vision. Thus, individuals experience an interactive three-dimensional environment, which gives them the opportunity to perceive and manipulate it as if it were a real one, while interaction with true reality is blocked. The power of VR comes from the fact that the human mind actually comes to believe that they are in that conjuncture, and all psychological phenomena – memorization, learning, feeling emotions, etc. – are experienced as if they came from the real environment (Pottle , 2019). Therefore, the use of this technology has begun to gain momentum in more and more fields, such as education, health, rehabilitation, professional training, or games.

Virtual reality simulation has also proven valuable in psychology, both for the treatment and cure of certain disorders, and for testing. A meta-analysis by Kim et al. in 2021 explored the development of VR technology over the past 20 years in neuropsychological testing. One important finding was that over time, ecological validity—the extent to which a test’s measurements can generalize to real-life behavior—has increased as testing frameworks have improved. This is quite intuitive, as VR is designed to make participants behave exactly as they would in real life, with much greater experimental manipulation. There has also been a shift from focusing on simpler psychological phenomena such as attention, inhibition or working memory to more complex ones such as decision-making, problem-solving or planning, which is a positive point as it means that the accuracy of the testing remains even if the testing becomes more complicated. In addition, this is also due to the reduction of the feeling of motion sickness induced by virtual environments, by adapting and perfecting these technologies over time, which also leads to the extension of the age range in which these tests can be applied, even to the elderly. On the other hand, when it comes to structural validity – whether what the test measures really reflects the theoretical model of the phenomenon it claims to measure – it can suffer when traditional tests are translated into VR. The reason is that this setting can introduce new variables that influence the testing, such as the attention that the participants pay or how present they feel, which can sometimes lead to distortion of the results. The study thus highlights the need to pay

increased attention to these aspects when creating any kind of test in VR, but it does not mean that this type of testing does not still have many benefits. Thus, the conclusion of the meta-analysis is that the use of this technology has increased greatly in recent years, becoming a promising and flexible framework for psychological assessment, predicting that it will become increasingly popular, specifying that the validation and optimization of VR instruments is a continuous necessity so that they become fully reliable and widely accepted (Kim et al., 2021). Other research has also had positive results regarding ecological validity. For example, in the context of diagnosing disorders defined by attention deficit such as ADHD , a main problem was this lack of exact overlap between the tasks in the tests and those encountered by individuals in real life. However, the VR versions are characterized by a stronger dynamism and interaction, which increases their validity (Rodríguez et al., 2018). In addition, the level of validity was also higher in the case of the method entitled "Virtual Reality Everyday Assessment Lab (VR-EAL)", in which participants perform everyday tasks, similar to those in everyday life, in an environment that is easy to manipulate by researchers so as to measure exactly what they need, virtual reality thus bringing a great plus in the context of psychological testing of this kind (Kourtesis et al., 2020).

Given the above, it is quite intuitive that the introduction of VR technologies has also been made within the framework of personality testing, the approaches and results being different from case to case. A first example had an approach quite similar to the present study, with participants completing a personality questionnaire in two phases, once in a virtual environment, after having the opportunity to create an avatar to their liking, and then, at a distance of 6 months, a traditional one, on paper. The researchers' hypothesis is that people are more authentic in virtual environments due to the feeling of anonymity, being able to be a closer version of the ideal self they have in mind, having a higher tendency to have the behaviors they want, but which they do not have the courage to expose in real life, perhaps because of the consequences (Aas et al., 201 cf. Wigboldus , 2008). The results were not as expected, however, with the differences between the two types of testing being insignificant; It should be noted that fidelity was ensured through a test that involved checking the inconsistency of the answers (Aas et al., 2010). Another personality assessment method was created by the Austrian company PwC with the aim of providing a more objective test of potential employees and which more accurately predicts the alignment of candidates' values with the organizational culture. Based on the HEXACO model, they created several mini-games in VR that aimed to assess the facets of the model. To validate the created assessment, participants also completed the traditional questionnaire, together with a test that refers to a person's resistance to change, to which 6 interview-type questions were added that were based on the company's values. The correspondence between the two types of assessment was relatively good, but the VR version found only five factors instead of six, as the traditional model has, and the structural fidelity was affected. In addition, a more detailed analysis was described in the article for Agreeableness and Openness to Experiences, which correlated with specific actions and choices in the game. At the same time, the experience was perceived in an exclusively positive way by all participants, regardless of whether they had had contact with VR technology before or not, but an important limitation to consider in interpreting all the results is the low number of participants and the lack of cultural diversity, but this innovation brought by the PwC company remains valuable for the intersection of the worlds of technology and psychology (Nickel et al., 2024). Another study used a similar idea of mini-games in VR that were built taking into account the Big Five model, and in addition to tracking the way they made decisions, the researchers also collected data on the participants' eye movements. Based on this data, they aimed to predict the participants' score on the NEO-FFI questionnaire (applied as the first stage of the experiment) by using Machine Learning tools. Learning . The accuracy of the prediction was quite high, varying from one

facet to another, but having an average of 78%. Thus, this type of procedures has proven to be valuable for the assessment of personality traits, the authors pointing out the benefits of future research in this area for the field of psychological testing and recruitment (Vargas et al., 2024).

Therefore, two of the main hypotheses related to why psychological testing in VR is worth exploring and may yield better results than traditional testing are ecological validity and increased participant engagement due to the sense of presence (as well as greater flexibility in item design). The way in which ecological validity is influenced in psychological testing in VR was extensively analyzed in an article published by Parsons and Phillips in 2016. They emphasize that the psychometric properties of assessments influence this validity, and new VR technologies have led to the development of accessible and easy-to-use systems that facilitate the collection of data related to an individual's functioning in real life. When it comes to the impact of the level of engagement in the virtual reality environment, a study conducted by Markowitz et al. in 2018 concluded that the level of knowledge accumulated by study participants was positively correlated with increased interactivity and more active exploration of the simulated reality. The credibility of the environment also has a strong influence in generating the feeling of social presence, which refers to the psychological state felt when an individual is with other people. The realism and credibility of agents in the virtual environment were measured using the Multimodal Presence Scale, and the results show that these dimensions significantly influence the emotional experiences of the participants. Thus, the more present an individual feels in the virtual environment, the more valuable the data obtained are for understanding and intervening in their behavior (Pfaller et al., 2021). However, there are also studies that have looked at the level of absorption and presence of individuals within a simulated VR environment as moderators on personality measurement and concluded that there is no significant relationship between these phenomena (Aas et al., 2010).

The relationship between empathy and virtual reality

The relationship between empathy and VR experience is a topic that has been the subject of much research, but many questions remain unanswered. A meta-analysis conducted by Martingano et al. in 2021 analyzed the effect of VR experience on the two dimensions of empathy – affective versus cognitive. The results showed that there is a significant effect on the first facet, this technology having the ability to arouse empathic feelings in individuals, but there was no significant relationship with cognitive empathy, suggesting that understanding the mental states of others was not amplified by the VR experience. The authors' hypotheses regarding this unexpected result are either that the emotions and thoughts of the characters are very explicit, the participant no longer having to make the effort to understand it himself, or because this context is over-stimulating, which limits the mental processing that would be necessary to exercise this type of empathy. In contrast, another study found a significant relationship between VR experience and cognitive empathy in a virtual reality perspective-taking (VRPT) experiment. The authors found that immersing participants in realistic scenarios facilitated emotional understanding and increased empathy with the character being portrayed, which was interestingly observed in individuals with lower levels of general empathy (Loon et al., 2018). Similar to the previous study, a common approach to analyzing the relationship between empathy and VR is to consider how the task of taking a character’s perspective influences the experience. Another study shifted the interpretation of empathy to the other two dimensions—state and trait—in the context of this change of perspective (Change of Perspective – CoP). Empathy as a state increased significantly following this procedure carried out in VR, especially in students,

compared to police officers, but this relationship was not moderated by the level of empathy as a trait, thus this dimension had no significant relationship either with empathy as a state taken in isolation, or in interaction with the perspective in VR. Instead, the increase in the feeling of immediate empathy could be interpreted as a result of the mechanisms of immersion and presence identified within VR (Teernstra , 2023). Similar results were also found in a study that measured empathy as a state in an objective way, after measuring it through a self-assessment questionnaire of empathy as a trait. Using EEG signals, it was found that people with a high level of empathy as a trait showed frontal alpha asymmetry in the brain even before being exposed to any empathetic stimulus. Following the VR experience, which involved viewing a domestic abuse of a child, significant physiological changes were observed indicating an affective reaction specific to empathy, which can be considered as a method of measuring empathy as a state. However, the level of empathy as a trait was not significantly related to these physiological reactions here either, highlighting the lack of correlation between the two dimensions of empathy (Alimardani et al., 2020). In addition, the authors tried to achieve a more pronounced state of immersion, as previous studies have shown that this also leads to a stronger sense of presence, which is correlated with a more intense emotional experience within an affective simulation (Alimardani et al., 2020 cf. Diemer et al., 2015).

Empathy, presence, and VR experience have been shown to be closely related in several studies. For example, in a simulation that involved putting homeless people in perspective, participants who reported feeling more connected to them also reported higher levels of empathy, but also scored higher on how present they felt with the VR characters. Thus, self-rated empathy was found to be significantly related to perceived personal discomfort, but also to social presence, which was also marginally significant in VR compared to the simulated version on a classic computer (Herrera et al., 2018). Similar results were found in another study, which found that the experience of a simulated reality led to greater involvement and a higher level of empathy for a refugee girl, compared to a control group. The authors state that a more pronounced engagement leads to a state of presence, which also includes a sense of reality towards what is experienced, this phenomenon can be seen as the connection between empathy and virtual reality (Schutte & Stilinović , 2017). Another study analyzed a multifactorial model involving empathy, a sense of presence, engagement, “ flow ” (a state of total involvement) and a sense of embodiment (“ embodiment ”). The author had five hypotheses that proved statistically significant, from which it follows that a sense of presence positively influences embodiment, a state of flow and empathy, and engagement is favored by a high level of embodiment and empathy (Shin , 2018). Thus, this study highlights the complex nature of the relationships between certain psychological phenomena, including empathy, within the experience of individuals in virtual reality, emphasizing the need for diverse approaches and future research in this area.

Since the relationship between empathy as a state and personality assessment is not deeply analyzed in the specialized literature, the present study has this aspect as its main objective. Taking into account the above, the way in which feelings of presence, commitment and reality have been proven to positively influence psychological assessment, and at the same time, concluding from the analyzed studies that they are in a positive significant relationship with empathy, the following hypotheses are proposed in this study.

Research hypotheses

H1: Empathy as a state and its dimensions measured after testing in the VR environment will have a higher level than those measured after assessment with non-verbal items.

H2: The level of empathy as a felt state and its dimensions is positively correlated with the level of empathy as a general trait.
H3: Rating fidelity (measured by internal consistency) is higher in the group of participants with scores above the median on Empathy as a state, compared to the group in which participants have scores below the median.
H4: Compared to empathy as an affective and associative state, cognitive empathy shows a stronger association with differences between the two personality measurements.
H5: The facets Altruism, Conformity, Sympathy Psychopathy (Agreeability), Sociability, Dominance, Stimulation Seeking (Extraversion), Openness to Feelings, Openness to Action (Openness to Experiences) correlate statistically significantly and positively with empathy as a state.
H6: Clinical facets (Negative Affectivity, Antagonism, Disinhibition) correlate statistically significantly, but negatively, with empathy as a state.

METHODOLOGY

Participant characteristics. Sampling procedures

The participants in the study were mainly students, the vast majority from the Faculty of Psychology and Educational Sciences, University of Bucharest, all of whom enrolled in the study on a voluntary basis, and their participation was conditional on completing a consent form, which specified the stages and objectives of the study. The only eligibility, inclusion and exclusion criteria were that the individuals were of legal age. Therefore, sampling was done by convenience and volunteerism.

Procedure

The study was divided into three phases. The first phase took place physically, in the VR laboratory at the Faculty of Psychology and Educational Sciences, University of Bucharest, coordinated by members of the Assessment and Individual Differences Laboratory (AID Lab , within the same faculty) and by student volunteers. The first step was the participants completing the registration and consent form, followed by instructions on how to use the equipment, setting up the VR headset, and finally, administering the test, which lasted an average of 30 minutes and included 126 items. The test itself consisted of observing situations illustrated by visual and auditory stimuli, and then selecting, using controllers, on a seven-way scale, how much they identified with what was presented (in addition, in situations where there were multiple characters involved, the one they had to relate to had a red triangle above it). After this assessment, a 20-item cognitive skills test followed, also conducted in VR, consisting of choosing from four 2D shapes, the correspondence for a 3D shape. The last part of this stage consisted of completing a form in Google Forms , consisting of the Empathy as a State Scale (Shen , 2010), which specified the need to relate to the characters and the experience in VR, and the Prysma Empathy as a General Trait Scale. The second and third stages were subsequently conducted online, with participants receiving the related forms by email. These were made up of non-verbal items based on exactly the same questions that were used to design the VR ones, plus again the Empathy as a State Scale, but with reference to the characters in the non-verbal items - the second stage -, respectively the same questions but in their original form - the third stage.

Tools used

Empathy as a State Scale

Empathy as a state was measured using a scale created by Shen in 2010, in response to the lack of a measurement instrument that separates this concept from empathy as a general trait and that meets the conditions according to Hunter and Gerbing (1982) – content validity (how the items reflect the theoretical construct), internal consistency (coherence of responses between items), and external consistency (relationship with other variables that could have an influence). The scale was validated in the context of message processing, empathy as a state being defined here as “a process in which the perception of the characters’ state automatically activates in the receiver an indirect (by substitution) experience of their state, situation, and goals”, a formulation that fits well in the context of measurement in the VR environment. In two studies conducted, with a total of N = 478 participants, this instrument demonstrated good internal and external consistency, convergent and discriminant validity, as well as good fidelity. The scale has 12 items, divided into 3 types of empathy as a

state: affective, cognitive and associative, originally in English, being translated by me into Romanian. An example of an item from each category is as follows: “I was in a similar emotional state to the characters during the testing.”, “I could understand what the characters were going through during the testing.”, “I could resonate with the characters’ experiences during the testing.”, with participants having to choose, on a Likert -type scale from 1 to 5, how much they find themselves in the respective statement.

Empathy Scale as a General Trait

The instrument used to measure empathy as a general trait is a scale from the Prysma manual , created by Iliescu and Ion (Prysma). Assessments , nd .). Although it has only 6 items, the scale has been empirically validated on several Romanian samples, being appropriate in the context of the present study, in which participants already have many stages to complete, avoiding additional fatigue. The items were also scored on a Likert -type scale from 1 to 5, an example of a statement being “I can easily sense the emotional atmosphere within a group.”.

Personality scales. Moving into VR. Non-verbal items

Since the main objective of this study was to build a personality questionnaire in the virtual reality environment, the scale that measures personality is new, but it is based on classic scales, deeply empirically validated. Thus, the researchers who worked on the foundation of this questionnaire used as a theoretical basis the Big Five Personality Model (the Five Factor Personality Model (FFM) and the Alternative Model of Personality Disorders (the Alternative Model of Personality Disorders – AMPD). The items used to create the VR scenarios were selected from the Prysma scales (Prysma Assessments , nd .) based on two main criteria: empirical validity (evidence from the literature that the measured trait is relevant for predicting relevant outcomes and behaviors in the workplace) and the possibility of adaptation to VR (the practical aspects of implementing the concepts in this space and whether they can be simulated effectively, while maintaining their validity). Taking these principles into account, the sub-dimensions that make up these theoretical models were analyzed and only those that meet both criteria were selected – examples: Altruism, Cooperation, Self-discipline, Adventurous Spirit, Perseverance (FFM); Hostility, Risk-Taking, Callousness (AMPD). The subsequent generation of scenarios, both through visual and auditory stimuli, was done with the help of artificial intelligence tools. The same items were used both to design the questionnaire in the second stage, of non-verbal personality items, also using artificial intelligence to generate the drawings, and for the third stage, where the items were directly in text form.

result

All statistical analyses were performed using the R Studio program.

Data normality analysis

Before running the analyses for hypothesis testing, a normality analysis of the data was needed to see if the tests for parametric data were viable. Thus, through the analysis of the mean and standard deviation, as well as the skewness and kurtosis methods , an approximate normality of the data measured for each facet, within all the assessment stages, could be noted. The same tests were also done for the data collected in Stages 0 and 1 for empathy as a state, on each dimension, but also for the data measured for empathy as a trait, all having a distribution quite close to normality so that the parametric tests were appropriate.

Internal consistency analysis

In addition, we conducted internal consistency analyses for each facet, measured at each of the three stages, to see how the fidelity of the assessment of each personality dimension evolves across the three forms. There is consistency for some facets – Sociability, Stimulus Seeking, Anxiety, Anger and Irritability, Impulse Control, but for the rest, a decrease in internal consistency is observed in Stages 0 and 1, compared to Stage 2. In the case of the last six facets, in the clinical category, internal consistency is negative for all stages, denoting a problem at the scale level and inconsistency between items. Therefore, the results presented below should be understood and used further taking this aspect into account.

Table 1. *Cronbach's alpha for the 3 stages*

Facet	α (Stage 0)	α (Stage 1)	α (Stage 2)
Altruism	.48	.42	.74
Conformism	.23	.18	.78
Sympathy_Psychopathy	.58	.60	.74
Ambition	.39	.44	.76
Self-discipline	.70	.05	.89
Perseverance	.24	-.05	.69
Sociability	.70	.71	.75
Dominance	.52	.20	.83
Stimulation_seeking	.73	.72	.88
Anxiety	.70	.65	.91
Anger_Irritability	.75	.71	.94
Impulse_control	.76	.79	.92
Vulnerability_to_stress	.66	.30	.87
Opening_feelings	.30	.60	.77
Open_action	.43	.18	.57
Emotional lability	-.87	-.67	-1.28
Hostility	-.36	-.96	-.58

Facet	α (Stage 0)	α (Stage 1)	α (Stage 2)
Manipulation	-.34	-.67	-2.00
insensibility	-.34	.01	-.45
Impulsivity	-.35	.04	-2.70
Risk taking	-.47	-1.30	-1.47

CFA Model Analysis

To verify whether the theoretical model for measuring Empathy as a state (affective + cognitive + associative) fits the empirical data collected in both stages, we conducted a confirmatory factor analysis , using the " lavaan " package (Rosseel , 2012). For interpretation, we extracted the adjustment indicators of the model. The results show a partially satisfactory fit: CFI = .91 > .90, TLI = .84, between .80 and .90, means a good, respectively moderate fit, SRMR = .05 < .08 also indicates a good agreement between the model and the data, while a problem appears at RMSEA = .16 which should have been at least < .08, an indicator denoting a poor fit.

In addition, we analyzed the standardized factor loadings of each dimension on the latent factor - the total score measured for Empathy as a state -, separated by stages (Table 2.). This provides a clear picture of the contribution of each facet, relative to the general construct, allowing us to see whether the proposed factor structure is valid and allows a valid interpretation of the differences and similarities between the two stages. All three dimensions have a substantial contribution to the latent factor in both stages, the largest being from associative empathy, which has a very good loading, especially in stage 1.

At the same time, we calculated Cronbach's alpha for the Empathy as a state scale in both stages, the results demonstrating excellent internal consistency: .89 for the data from Stage 0 and .90 in Stage 1. We conducted the same type of analysis for the Empathy as a trait scale, resulting in alpha = .73, thus an acceptable internal consistency.

Table 2. *Standardized factor loadings*

left	RHS	east	std.all
Empathy_state_total_stage0	Affective_empathy0	2.51	.66
Empathy_state_total_stage0	Cognitive_empathy0	2.26	.74
Empathy_state_total_stage0	Associative_empathy0	2.70	.85
Empathy_state_total_stage1	Affective_empathy1	2.80	.74
Empathy_state_total_stage1	Cognitive_empathy1	2.57	.71
Empathy_state_total_stage1	Associative_empathy1	3.10	.96

Comparison of Empathy as a State in the Two Stages

Initially, to ensure that I could also perform parametric tests in this case, I used the Shapiro Test to verify that the differences between the scores measured for Empathy as a state in the two stages have a normal distribution, which turned out to be true. This allowed me to continue using the paired samples t-test . samples t-test) to evaluate the difference between the two measurements of Empathy as a state and see if it is significant or not, on each dimension.

Table 3. *t-test between Empathy as a state measured in Stage 0 and Stage 1*

vary	t	df	p	Mean diff	BUT_ lower	BUT_ upper	Interpretation
Empathy affect	1.18	109	.23	.44	-.30	1.19	insignificant
Empathy cognition	5.08	109	.00	1.86	1.13	2.59	significant positive
Empathy association	5.58	109	.00	1.76	1.13	2.39	significant positive
Empathy total state	4.58	109	.00	4.07	2.31	5.83	significant positive

Empathy as a state was on average significantly higher in Stage 0 compared to Stage 1, which was observed for both the total score and the cognitive and associative dimensions, validating Hypothesis 2. In the case of affective empathy as a state, the difference turns out to be not significant.

Correlations Empathy as a state and Empathy as a general trait. Correlations between dimensions of Empathy as a state

Pearson correlation test between the total score obtained for Empathy as a state, as well as the scores obtained on each facet, at both stages, and the score obtained when measuring Empathy as a general trait. An interesting result was obtained, the correlation being significant (positive) only for Empathy as a state measured at stage 0. Therefore, Hypothesis 3 is only half validated, but we can speculate that the lack of correlation in the case of the measurement at stage 1 is related to the lower level of Empathy as a state at this stage, compared to stage 0, as we saw in the previous analysis.

In addition, we also conducted Pearson correlation tests between the dimensions within Empathy as a state – only between those that were part of the same stage -, the results being significantly positive in each case, with $p < 0.01$, so the correlation is high.

Table 4. *Correlations between the dimensions of Empathy as a state Empathy as a general trait.*

Variable 1	Variable 2	R	p	Interpretation
Affective_empathy0	Empathy_general_trait	.21	.02	significant positive
Affective_empathy1	Empathy_general_trait	.11	.23	insignificant
Cognitive_empathy0	Empathy_general_trait	.19	.04	significant positive
Cognitive_empathy1	Empathy_general_trait	-.04	.63	insignificant
Associative_empathy0	Empathy_general_trait	.29	.00	significant positive
Associative_empathy1	Empathy_general_trait	.07	.47	insignificant
Empathy_state_total0	Empathy_general_trait	.27	.00	significant positive
Empathy_state_total1	Empathy_general_trait	.05	.58	insignificant

Variable 1	Variable 2	R	p	Interpretation
Affective_empathy0	Cognitive_empathy0	.50	.00	significant positive
Affective_empathy0	Associative_empathy0	.55	.00	significant positive
Affective_empathy1	Cognitive_empathy1	.54	.00	significant positive
Affective_empathy1	Associative_empathy1	.71	.00	significant positive
Cognitive_empathy0	Associative_empathy0	.64	.00	significant positive
Cognitive_empathy1	Associative_empathy1	.68	.00	significant positive

Group division by median Empathy as a state. Internal consistency comparison

To test Hypothesis 3, I divided the group into two, taking into account the median of Empathy as a state, and then calculated Cronbach alpha and see if there is a significant difference between the groups (the hypothesis claiming that in the group with Empathy as a higher state, the value calculated for internal consistency should have had higher values than for the other group). The hypothesis is not validated, however, for most facets the difference is insignificant, the only facets where the result is significantly positive being Stimulation Seeking (division by total score and by cognitive dimension score, only in Stage 0) and Anger Irritability (only for cognitive dimension, Stage 0), and in the case of the facets Openness to Action (for total score, Stage 0) and Sympathy Psychopathy (for cognitive dimension, Stage 0), the result is significant, but reversed. Although I conducted this analysis for the division by all 8 constructs, it was redundant to attach all the tables, since in all other cases, in which the division was made by the other six constructs, all the differences were insignificant. Thus, Hypothesis 3 cannot be validated.

Table 5. *Cornbach's alpha calculated for two groups (low vs. high), divided by the median of the variable Empathy_state_total0*

Facet	low_alpha	high_alpha	Interpretation	p
Altruism	.47	.59	insignificant	.32
Conformism	.33	.19	insignificant	.48
Sympathy_Psychopathy	.57	.54	insignificant	.79
Ambition	.22	.46	insignificant	.16
Self-discipline	.67	.70	insignificant	.73
Perseverance	-.03	.32	disabled alpha	N/A
Sociability	.74	.75	insignificant	.79
Dominance	.55	.48	insignificant	.58
Stimulation_seeking	.66	.82	high > low	.02
Anxiety	.73	.56	insignificant	.07
Anger_Irritability	.67	.77	insignificant	.19
Impulse_control	.76	.77	insignificant	.90
Vulnerability_to_stress	.72	.56	insignificant	.09
Opening_feelings	.30	.19	insignificant	.60
Open_action	.59	.13	low > high	.00
Emotional lability	-1.06	-1.47	disabled alpha	N/A

Facet	low_alpha	high_alpha	Interpretation	p
Hostility	-.33	-.39	disabled alpha	N/A
Manipulation	-.83	-.49	disabled alpha	N/A
insensibility	-1.40	-.21	disabled alpha	N/A
Impulsivity	-.34	-1.09	disabled alpha	N/A
Risk taking	-.81	-.86	disabled alpha	N/A

Table 6. *Cornbach's alpha calculated for two groups (low vs. high), divided by the median of the variable Empathy_cognitiva0*

Facet	low_alpha	high_alpha	Interpretation	p
Altruism	.53	.56	insignificant	.75
Conformism	.23	.30	insignificant	.71
Sympathy_Psychopathy	.67	.32	low > high	.01
Ambition	.28	.42	insignificant	.41
Self-discipline	.64	.71	insignificant	.42
Perseverance	-.02	.34	disabled alpha	N/A
Sociability	.77	.71	insignificant	.41
Dominance	.49	.55	insignificant	.66
Stimulation_seeking	.68	.82	high > low	.02
Anxiety	.70	.61	insignificant	.36
Anger_Irritability	.63	.81	high > low	.01
Impulse_control	.75	.76	insignificant	.80
Vulnerability_to_stress	.70	.59	insignificant	.29
Opening_feelings	.36	.05	insignificant	.17
Open_action	.50	.31	insignificant	.24
Emotional lability	-.68	-2.44	disabled alpha	N/A
Hostility	-.36	-.34	disabled alpha	N/A
Manipulation	-.93	-.45	disabled alpha	N/A
insensibility	-1.61	-.07	disabled alpha	N/A
Impulsivity	-.45	-.94	disabled alpha	N/A
Risk taking	-.60	-1.19	disabled alpha	N/A

Correlation between scores differences from Stages 0, 1 and Empathy as a state scores from Stages 0, 1

Hypothesis 4 was tested by analyzing the correlation between a “consistency” score between Stages 0 and 1, defined by the sum of the differences (in mode) between the scores obtained in the two stages, on each facet. The difference between the scores obtained in each stage, on each dimension of Empathy as a state, was also calculated. Then, by means of Pearson correlation tests , we verified whether this “consistency” score is significantly correlated and in what sense (the hypothesis being supported if the sense is positive) with the

difference between Empathy as a state felt by the participants. The correlation is indeed significantly positive only in the case of Cognitive Empathy as a state. Therefore, we can validate Hypothesis 4, with the mention that it also stands up by comparison with the total score measured.

Table 7. *Differences table*

Type_Empathy	R	p	Significant
Empathy_state_total	.14	.14	insignificant
Affective_empathy	.08	.39	insignificant
Associative_empathy	.02	.82	insignificant
Cognitive_empathy	.22	.01	significantly positive

Correlations between empathy dimensions and personality facets

To analyze the last two hypotheses, we applied the Pearson correlation test between all dimensions of Empathy as a state and all facets of personality (preserving the correspondence between the stages). In Table 8. There are only the pairs that had a significant correlation, the entire table having 209 rows. It can be seen that the facets Self-Discipline, Altruism and Openness (FFM) to feelings correlate positively with at least one dimension of Empathy as a state. Negative Affectivity and Disinhibition (clinical dimensions) also correlate positively, contrary to Hypothesis 6. In contrast, the Openness to action facet correlates both positively and negatively with the dimensions of Empathy as a state (total, stage 1, respectively associative, stage 0). At the same time, Antagonism correlates negatively with 2 of the dimensions measured in stage 0 (cognitive, associative), but also with the total. The conclusion is that the relationship between the dimensions of Empathy as a state and each facet of personality is complex, not being able to absolutely confirm any of the hypotheses (5 and 6), but the relationships found are worth remembering - a more detailed analysis will follow in the Discussion section.

In addition, we conducted a more detailed analysis of the relationship of each facet with the dimensions of empathy, to see if there are significant differences in the level of correlation – that is, if any dimension is significantly more correlated than another in relation to a certain facet; for this analysis we used the Steiger Test , 1980 for dependent correlations with a common variable. The cases in which the difference was significant appear in Table 9, being only 5 out of 108, in 4 out of the 5, Cognitive Empathy being the most correlated with the facet in question. Only for Conformity, in stage 0, the significant difference appeared between affective and associative. Thus, we cannot conclude that in general there are differences in the level of correlation, but those that proved to be significant are to be taken into account – especially since it was mostly in the case of Cognitive Empathy as a state, which was the only one with significant results and in the case of the correlation between the differences in the scores obtained; In addition, also mostly – 4 out of 5 -, the differences appeared in the case of the facets measured in Stage 0.

Table 8. *Correlations between dimensions of Empathy as a state and personality facets (grouped by stages)*

Empathy	Facet	R	p	Interpretation
Empathy_ cognitive0	Opening_feelings0	.19	.04	significantly positive

Empathy	Facet	R	p	Interpretation
Empathy_ cognitive0	Antagonism0	-.23	.01	significantly negative
Empathy_ associative0	Opening_action0	-.27	.00	significantly negative
Empathy_ associative0	Antagonism0	-.31	.00	significantly negative
Empathy_ associative0	Disinhibition0	.26	.00	significantly positive
Empathy_ status_total0	Antagonism0	-.27	.00	significantly negative
Empathy_ status_total0	Disinhibition0	.21	.02	significantly positive
Empathy_ emotional1	Self-discipline1	.22	.02	significantly positive
Empathy_ cognitive1	Altruism1	.19	.04	significantly positive
Empathy_ associative1	Negative_affectivity1	.20	.03	significantly positive
Empathy_ status_total1	Self-discipline1	.19	.04	significantly positive
Empathy_ status_total1	Opening_action1	.18	.05	significantly positive

Table 9. *Significance of differences Correlations*

Facet	Comparison	p	Meaning
Conformity0	Cognitive versus Associative	.00	significant
Conformity0	Affective versus Associative	.02	significant
Ambition0	Cognitive versus Associative	.00	significant
Open_action0	Cognitive versus Associative	.00	significant
Antagonism1	Cognitive versus Affective	.04	significant

DISCUSSIONS

Main results

The central variable analyzed in this study was Empathy as a state, measured in two different contexts to see if there are different manifestations, and how they affect the personality assessment. A first important result obtained was that the average difference between the scores obtained in the two stages is significant, the VR environment evoking a higher level of Empathy as a state than the illustrations. The result obtained makes sense, due to the feeling of immersion generated by VR, the situations depicted being much closer to everyday reality (Pottle , 2019), compared to the illustrations in Stage 2. Although following the searches carried out in the specialized literature we did not find a study that exactly follows this difference, there are several studies that affirm a high level of empathy as a state in the context of virtual reality (Teernstra , 2023 ; Martingano et al., 2021 ; Shin , 2018). Moreover, this difference is observed only for the cognitive and associative dimensions, not for the affective one. This means that there is a real difference between these types of empathy and that certain forms have a greater influence depending on the context. Taking into account the nature of the test and the fact that the intention was not to arouse strong feelings, but to understand the situations presented, either objectively, by understanding the thoughts, emotions and attitudes of another without appropriating them – the cognitive dimension - or by making associations between what the individual saw and what he had experienced in the past – the associative dimension. In the literature there are contradictory conclusions to each other when it comes to the differences between the dimensions of empathy in VR, Martingano et al., 2021 highlighting the higher effect of the affective one, while the study conducted by Loon et al. in 2018 demonstrated the increase in cognitive empathy. This fact denotes that empathy as a state is indeed greatly influenced by context, but in this study the result obtained is consistent with the generated environment, with the way the evaluation method was constructed, and with the goal it wanted to achieve.

Another important conclusion that can be drawn from this study is the significant correlation between all dimensions of Empathy as a state and Empathy as a general trait, only in the case of the VR measurement, in relation to Stage 1 all correlations being insignificant. The result is not necessarily in contradiction with the specialized literature, as there are so many studies that have found a significant connection between the two, even at the level of separate analysis by dimensions (Wood et al., 2013 cf. Rusting , 1999; Van der Graaff et al., 2016, cf. Batson et al., 1986; Westbury and Neumann, 2008), as well as studies that concluded the opposite, even in the VR context (Teernstra , 2023; Alimardani et al., 2020). This analysis once again emphasizes the importance of the environment for this psychological phenomenon, even in the interaction with a variable to which the existence of a connection can be intuitively predicted.

Interpretation of results

At the level of fidelity of the two forms of personality assessment, it can be observed that it can vary quite a lot from one facet to another, as well as from one stage to another. Thus, for the facets Sociability, Stimulation Seeking, Anxiety, Anger and Irritability, Impulse Control, Sympathy and Psychopathy, the internal consistency is maintained around an acceptable or moderately acceptable value for both stages. In contrast, for Self-Discipline, Vulnerability to Stress, the values are reasonable only in the case of VR testing, and for Openness to Feelings it is the opposite, only in the case of Stage 2. Otherwise, the rest of the facets have an internal consistency below the acceptance threshold, and in the case of clinical

dimensions, the value is even negative. Therefore, the findings must be treated taking this aspect into account, for facets in which even one of the values is below the threshold, the results need to be adopted with some caution.

The influence of Empathy as a state on the two types of evaluation is another point reached in the context of this study, being analyzed in two ways. When it comes to the difference that appears at the level of internal consistency depending on how high the empathy as a state was, the results were insignificant for most facets. If we focus on the cases in which there was still a difference, we can see that they are only part of Stage 0, by dividing by the total score calculated or by the cognitive dimension, which highlights again the greater influence of empathy as a state in the VR context, compared to the one with illustrations, but also the high importance of cognitive empathy, which best fits the way in which the participants had to approach the items. Thus, upon closer analysis, in all three cases where the difference was significantly greater for the group with high Empathy as a state – the facets being Stimulation Seeking in both cases and Anger Irritability for the division by cognitive dimension -, the internal consistency exceeded the threshold for which reliability is very good. On the other hand, in the two cases where the difference was significant, but in the opposite direction – the facets being Openness to Action, Sympathy Psychopathy -, the internal consistency for the group with low Empathy as a state remained at a level that denotes moderate reliability. Therefore, it can be concluded that for certain facets, the high level of Empathy as a state positively influenced the quality of the assessment, reaching values higher than all the values initially calculated for internal consistency (before the division). Although no robust conclusion can be drawn from these analyses, it is worth noting that Empathy as a state has a conclusive effect in the context of psychological assessment, and future studies can build on these results to build other research methods for this influence. We did not find other studies in the scientific literature that analyze this relationship to be able to compare the results.

The second method of investigation brought another addition regarding the high relevance of the cognitive dimension in Empathy as a state. By analyzing the differences obtained in both the facet scores and those of empathy as a state in the two stages, it was noted that the only dimension where the correlation is significant is the cognitive one, although in the case of the total score the correlation is insignificant. Thus, it can be concluded that there are notable differences in the level of how people understand and objectively interpret the attitudes and feelings expressed by others, depending on the environment in which they are asked to do so, and these differences are also reflected in the way they answer questions designed to measure personality constructs. Therefore, these results are conclusive in understanding the factors that can affect psychological assessment and how they can be manipulated for measurements that better reflect the inner reality of individuals.

To have an overview of the interaction between the facets and the dimensions of empathy as a state, in the last part we analyzed each such pair, per stage. The only facets in the FFM category that have a significant correlation are Openness to Feelings, Altruism – with the cognitive dimension – and Openness to Action, Self-Discipline – with the associative dimension (negative sense), respectively the affective dimension, and with the total score (both positive). Thus, it can be concluded that individuals with a high score on these facets – respectively low, for Openness to Action -, will tend to feel more empathy at the moment towards the characters in the situations presented. This seems intuitively correct, considering that Openness to Experience (Abramson et al., 2022) and Agreeableness (Taylor et al., 2010) are seen in close relationship with empathy as a general trait, but Conscientiousness was also found in the same situation, in several studies on medical students (Suh et al., 2019 ; Costa el al., 2014 ; Guilera et al., 2019); however, it is worth

noting that this relationship is also reflected for empathy as a state. In the case of Openness to Action, since both a positive and a negative relationship emerged, a clear conclusion cannot be reached, but this inconsistency could also be explained by the lack of obtaining high internal consistency, in both stages. On the clinical facets side, Antagonism correlated negatively with two dimensions (associative and cognitive), but also with the total score. This was expected given its relationship with empathy as a general trait, which is also negative according to the literature (Jenkins , 2019). Negative affectivity is positively related to associative empathy, which could be explained by the fact that individuals manage, by referring to their own previous experiences, to empathize, especially if these are in the negative spectrum of emotions. However, according to the literature, hostility, a component of this facet, is positively related to difficulties in emotional regulation and negatively related to perspective taking, with empathy playing a moderating role (Contardi et al., 2016), therefore contrary to the result of the current study. However, the emotions provoked by the situations presented in the assessment do not have a strong negative valence, so Negative affectivity may not have the same effect. Disinhibition was positively related to empathy as an associative state, but also to the total score. Previous research has found a negative relationship with trait empathy in the context of cyberbullying (Wright & Wachs , 2020), thus contradicting the current result, but another study demonstrated that a higher level of Disinhibition indicated greater empathy, but without the association being statistically significant (Pauli et al., 2021). The main problem, however, is that all three of these clinical facets obtained negative values in the internal consistency analysis, so the results related to them are questionable.

The analysis started at the end aims to investigate in more detail which dimension of empathy correlates more strongly with certain facets. The differences between the individual effects of the dimensions are not significant in many cases, thus denoting that the distribution is quite uniform. However, in cases where a difference is observed, the cognitive one turns out to be the one with the strongest effect, in the case of Conformity, Ambition, Openness to Action and Antagonism. This is not an unexpected result, considering that this dimension was the only one for which the differences between the stages were significant, highlighting instead even more the strong effect it has in the context of understanding the situations presented in the assessment. The only significant difference that did not involve this dimension was in the case of Conformity, where the link was stronger for affective empathy as a state, compared to the associative one. However, in all four facets mentioned here, internal consistency is below the acceptable threshold, so these conclusions should be used with caution, requiring future studies to increase robustness.

Theoretical implications

Taking into account the lack of rigorous scientific literature on the differentiation between Empathy as a state and as a general trait, but especially on the measurement of the two in different ways, this study contributes in this direction. Following the analysis of the CFA model for the two evaluation stages, it can be concluded that the division into three dimensions – affective, associative and cognitive – is appropriate and can be further used in practice. Since all of them had a high factor loading , each one must be kept when designing a measurement scale. In addition, there were significantly positive correlations between the three forms of empathy, in both stages, which supports the validity of the theoretical model, by highlighting a conceptual coherence – they are interconnected - and convergent validation – the measured aspects belong to the same theoretical concept. It is also noted that the strength of the correlations is not extremely high, so each of the three brings its specific contribution, these not being overlapping. The validity of the theoretical model is also

supported by the high internal consistencies calculated for the Empathy as a state scale, for the data sets from both stages.

Practical implications

Other future research can focus on creating forms of assessment in the VR environment that have a high practical utility. This aspect can be achieved by focusing on building tests that intentionally stimulate the individual empathically. Emphasis should be placed (something that is missing in the present study) on measuring the ecological validity, but also the validity as such of the assessment form - the ability to measure what it is theoretically supposed to measure - by manipulating other external factors or by tracking convergence with other external indicators. Following the results of this study, it is recommended to investigate the relationship between validity and empathy as a state.

At the same time, to further emphasize the practical applicability, these results can be transposed or replication can be attempted (with the above mentions) in other psychological contexts, not only that of personality assessment. There are already many areas in which the VR environment is useful, such as psychotherapy - treating phobias, addictions or anxiety disorders -, psychological and cognitive rehabilitation - in the case of diseases or following traumas -, educational psychology and professional training, clinical assessment. Introducing the analysis of the concept of empathy as a state and its intentional stimulation can have positive results in such areas as well, while also adding to the insufficiency of scientific research on this psychological phenomenon, the VR context proving to be a conducive environment for such an analysis.

limitation

A first limitation of the study is represented by a small sample size (especially the group that completed all three stages), which may affect the statistical power of the tests conducted and the stability of the estimates made through CFA or correlation comparisons. At the same time, the diversity of the sample represents a limit, most of the participants being students at the Faculty of Psychology, University of Bucharest. This aspect also leads to a limitation of the age range of the participants, mainly between 18 and 23 years old, with a few exceptions. Thus, the generalization of the results is limited by the strong similarities between the participants.

Another limitation is represented by the technical problems that arose during the experiment. Some participants had to retake the test after they had already completed part of it, due to problems with the system. Thus, some items were seen twice in a row, and there is a possibility that the answer was no longer as authentic. In addition, these problems can induce frustration in the individual, then rushing to finish, paying less attention to the items, or simply distorting the answers due to an unpleasant mood. In addition, the lack of familiarity with the VR environment for some participants may affect the quality of the results as they may no longer feel at ease, or it may take a while to get used to it, which can lead to inconsistency in the answers. Also, also related to the specifics of the VR experience, some people may be hurt during the assessment (Giri & Agrawal , 2020). Although a coefficient was also measured throughout the project based on how the participant felt from this point of view, this aspect was not taken into account in this study.

Due to the fact that the study is carried out in three stages, a feeling of fatigue/boredom may appear, especially since the same items are repeated, although in a different form – participants may not even realize that the assessment is designed based on the same questions, but the simple fact of having to complete three personality tests may

induce this feeling -, which may affect the quality of the measurements. In addition, being a form of self-report assessment, the bias of being socially desirable may appear – especially in the measurement of empathy, especially for that as a general trait -, or a complete misunderstanding of the items may occur.

In addition, the lack of a randomised control group may be a limitation. Given that the main variable is Empathy as a state and is measured in two different stages, such an approach would have been useful to clarify whether Empathy as a felt state influences participants and how they respond, or whether there are other contributing factors – such as the feeling of novelty brought by VR, the increased interest or attention generated by this environment. Thus, a strict isolation of the effect of this variable on the test forms is impossible. No randomisation related to the order in which the three stages are supported was done, which may affect the relationships between the measurements taken.

The results regarding internal consistency per facet were not as expected, which indicates a problem in the initial design of the tests. Therefore, the results should be interpreted with this aspect in mind. Also, the lack of more studies related to empathy as a state, especially in the context of VR, may lead to the lack of a robust theoretical model on which to base the research from the beginning – but this is exactly the aspect that we want to combat, by contributing to filling the gap in the literature in this field.

Future directions

A first aspect that should be addressed in future research would be the diversification of participants, in terms of background and age, which will be done in the continuation of the project of which this study is part.

When it comes to the research model, as I mentioned above, a stricter isolation of empathy as a state by introducing a control group, or another VR environment for personality assessment, but which is not based on the appearance of characters, so that the feeling of empathy is no longer aroused. In this way, the influence of empathy as a state on personality assessment can be more truthfully determined. In the same direction, future studies could consider measuring other aspects that theoretically differentiate the VR experience from classical methods: the feeling of presence, engagement, the state of total involvement (“ flow ”) or the feeling of embodiment (“ embodiment ”) (Shin , 2018), so that the individual impact of empathy as a state in this context can be highlighted. Following the conduct of this study, I believe that exploring the relationship between empathy as a state and engagement during test forms is an aspect that deserves more attention.

Given that the results were positive regarding obtaining a higher state empathy score in VR compared to the illustration method, it can be understood that more thorough testing of this psychological phenomenon in immersive experiences of this kind makes sense. Thus, future research could try experimentally manipulating this variable directly in the VR environment, intentionally evoking weaker or stronger emotional reactions, and analyzing how Empathy as a state is influenced or influences participants' responses to stimuli.

Conclusions

The present study makes a valuable contribution to the literature on empathy - especially as a state -, personality assessment, but especially the relationship between the two. Following the start of the statistical analyses, the main hypotheses were validated, but the results of the secondary hypotheses should also be taken into account. However, it is important to mention again the consideration of a level of reluctance for the analyses relating to the facets for which the internal consistency was below the acceptable threshold. Future

research is needed to lead to the robustness of the conclusions, but the present results provide a solid basis for further in-depth studies.

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